

A Three-Dimensional Grouting Model Considering Hydromechanical Coupling Based on the Combined Finite-Discrete Element Method

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Abstract: In this paper, we present a three-dimensional (3D) grouting model based on the combined finite-discrete element method (FDEM). The 3D grouting model discretizes the problem domain into tetrahedral elements and joint elements, and the grout flows only in the broken joint elements, which satisfies the planar Poiseuille flow. By combining grout migration, FDEM mechanical cracking computation, and stress-induced fracture aperture variation in a graphics processing unit (GPU) parallel multiphysics FDEM software, called MultiFracS, the 3D grouting model can model rock cracking and the effect of hydromechanical (HM) coupling. First, a grouting example with analytical solutions is presented to validate the 3D grouting model. Then, we investigate the influence of several key parameters on grout penetration in fractured rock masses. The results reveal that the 3D grouting model can model grout migration, pressure distribution, grout–rock mass interaction, rock deformation, and crack initiation and propagation. Finally, the evolution of fracture geometry induced by grouting under different in situ stresses is studied. The numerical results present high coincidence with the in situ experimental results, demonstrating that the 3D grouting model is effective in dealing with fracture grouting. DOI: [10.1061/\(ASCE\)GM.1943-5622.0002448](https://doi.org/10.1061/(ASCE)GM.1943-5622.0002448). © 2022 American Society of Civil Engineers.

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Introduction

Grouting is one of the most commonly used methods to deal with incomplete geological structures such as fractured fault zones in underground excavation. It can effectively improve the mechanical properties of fractured zones, reduce water inflow, and ensure the stability of rock walls during excavation (Gustafson and Stille 1996; Hässler et al. 1992; Yang et al. 2002). However, current theoretical research on grouting lags far behind engineering practice. It is necessary to study the migration mechanism of fracture grouting and develop appropriate grouting theories to guide grouting engineering practice.

In the past few decades, many research studies have been conducted on the grout flow mechanism in fractured rock masses. In terms of experiments (Funeag and Fransson 2006; Liang et al. 2019; Sui et al. 2015; Yang et al. 2016a, 2018a), Funeag and

Fransson (2006) designed field tests using silica sol as a grouting material to study the groutability of low-permeable rocks; Liang et al. (2019) carried out grouting tests in an inclined fracture with sand and water flow and studied the migration of chemical grout, and Sui et al. (2015) used a grouting system with a transparent fracture replica to conduct experimental research on the chemical grouting in a fracture with flowing water. However, it is still difficult to observe grout migration and fracture network evolution directly in the field.

Therefore, scholars (Amadei and Savage 2001; Funeag and Gustafson 2008; Funeag and Thörn 2018; Gustafson et al. 2013; Pedrotti et al. 2017; Zhang et al. 2018) have studied the grouting process from the perspective of analytical solutions. Funeag and Gustafson (2008) deduced analytical solutions of the penetration length of a Newtonian fluid in one-dimensional channel flow and two-dimensional radial flow. Gustafson and Stille (2005) obtained analytical solutions of the grout penetration length of Bingham fluid in one-dimensional channel flow and two-dimensional radial flow. However, these analytical solutions are usually applicable only to grouting in simple fractures.

With the rapid development of numerical calculation theories, a large number of numerical methods (Chen et al. 2014; Eriksson et al. 2000; Fidelibus and Lenti 2012; Kim et al. 2009; Sun et al. 2018) have been applied to the simulation of grout migration. For instance, Chen et al. (2014) established a numerical model for simulating the grouting process in fractures by combining the finite-element method and the finite volume method; Saeidi et al. (2013) used the Universal Distinct Element Code (UDEC) to study the effects of joint properties and grout properties on grout flow and penetration length; Mohajerani et al. (2015) developed an explicit grouting forehead pressure (EGFP) algorithm to study grout penetrability in fractured rock masses and extended the algorithm to two-dimensional discrete fracture networks (Mohajerani et al. 2017), and Zou et al. (2019) studied the impact of network structure and hydraulic variability on grout propagation using a two-phase flow model in saturated networks. Recently, the numerical manifold (NMM) has been widely used in geotechnical and

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geological engineering (Yang et al. 2019, 2020a, 2016b, 2020b). For example, Yang et al. (2018b) modeled hydraulic fracturing using the numerical manifold method.

However, these numerical models either do not consider crack initiation and propagation during grouting (Eriksson et al. 2000; Kim et al. 2018) or they do not consider the interaction of grout and rock masses (Mohajerani et al. 2015; Saeidi et al. 2013; Sun et al. 2019, 2018). During the grouting process, the grout pressure may significantly change the apertures of rock fractures, induce new cracks, etc. Therefore, considering these effects is important to realistically simulate the grouting process. Moreover, these models are primarily limited to two-dimensional cases. Three-dimensional grouting models are relatively uncommon due to the intrinsic difficulty in handling complex 3D fracture networks. Therefore, simplified methods, such as unified pipe-network, are developed (Sun et al. 2018). To overcome the drawbacks of previous models, in this study, we develop a three-dimensional grouting model based on the combined finite-discrete element method (FDEM), which is implemented in a GPU parallel multiphysics finite-discrete element software, namely Multi-FracS. The proposed 3D grouting model takes advantage of the FDEM's ability in simulating cracks in solid materials, and it can account for crack opening and closing, as well as crack initiation and propagation during the grouting process.

The FDEM was first proposed by Munjiza et al. (2015, 1995), Munjiza (2004), and Munjiza et al. (2011). It has been widely employed in recent years to solve challenges in rock mechanics and rock engineering (Lei et al. 2019; Lisjak et al. 2014, 2015, 2013; Mahabadi et al. 2012; Rougier et al. 2011, 2014; Wang et al. 2022a, b). The FDEM was originally used to simulate the fracture and fragment of solids. Subsequently, two-dimensional and three-dimensional hydromechanical coupling models based on the FDEM have been developed (Lisjak et al. 2017; Yan et al. 2021a, 2016, 2018; Yan and Zheng 2017a; Yan et al. 2022d, e), which allows the FDEM to solve fluid-driven fracturing problems. In addition, based on the newly developed hydro-thermo-mechanical model (Yan et al. 2022b), moisture-fracturing coupling model, and contact heat transfer model, the FDEM can also simulate thermal cracking of solid materials (Yan and Jiao 2020, Yan et al. 2019b, 2021d, e), the hydrothermal coupling effect in fractured rocks (Yan et al. 2019a, 2022c; Yan and Jiao 2019), soil shrinkage and cracking (Yan et al. 2021c, g, 2022c), and contact heat transfer, and induce its thermal cracking (Yan et al. 2022a, 2021f). Although the two-dimensional grouting model is proposed for fractured rock mass recently (Yan et al. 2021b), no three-dimensional FDEM-based grouting model for simulating grouting in fractured rocks has been reported.

The structure of this paper is as follows. The "Fundamentals of the Combined Finite-Discrete Element Method" section briefly introduces the basic principle of the 3D FDEM. In the "3D FDEM Grouting Model" section the 3D grout flow model is built using the 3D FDEM's unique topological relationship between joint elements and tetrahedral elements. Then, a 3D grouting model is constructed by combining the FDEM mechanical cracking calculation and the 3D grout flow model. The 3D grouting model can simulate grout migration, the deformation of rock masses, grout-rock mass interaction, and initiation and propagation of cracks. To demonstrate the correctness of the 3D grouting model, an example is presented in the "Model Verification" section. In the "Grouting Procedure Simulation in Rock Masses with Complex Crack Networks" section, the procedure of grouting in cracked rock with complex fracture networks is studied using the 3D grouting model. Besides, the influences of grout viscosity, yield stress, injection pressure, and fracture aperture on the maximum grout filling ratio are investigated, respectively. In the "Model Verification" to

"Simulation of Fracture Grouting under Different In Situ Stress" section, the evolution of geometric shapes of grouting fracture under different in situ stresses is studied. The numerical simulations are compared with the experimental results.

Fundamentals of the Combined Finite-Discrete Element Method

Basic Concept

The basic idea of the 3D FDEM is to divide the discontinuous domain into a tetrahedral finite-element mesh with the discontinuous surface as a constraint, as shown in Fig. 1(a). As shown in Fig. 1(b), a six-node joint element with zero initial thickness is introduced into the shared surface of adjacent tetrahedral components. The joint elements at the discontinuous surfaces are initially marked as broken joint elements to characterize pre-existing fractures. Other joint elements are labeled as unbroken joint elements in continuum medium.

The FDEM solves the state of tetrahedral elements (including stress and strain) using the finite-element approach, and models contact between tetrahedral elements using the discrete-element method. Deformable tetrahedral elements and bonding-effect joint elements can be used to model the deformation of solid materials. In addition, the FDEM can model crack initiation and propagation, as well as solid material fracture and fragmentation by the broken joint element. Many papers, including those of (Mahabadi et al. 2014; Tatone and Grasselli 2015; Yan and Zheng 2017b), have described the basic principles of the FDEM in detail.

Constitutive of Joint Element

The joint element is inserted between the adjacent tetrahedral element when the FDEM mesh is divided. Under the influence of external load, the two tetrahedral elements that are connected by the joint element can move to each other relatively. The relative motion can be divided into two components: normal displacement o and tangential displacement s , which are perpendicular to the joint plane and parallel to the joint plane, respectively. For the opening, the normal displacement is positive, and for closure, it is negative. The normal displacement o and tangential displacement s can be used to determine the damage state of the joint element. The critical

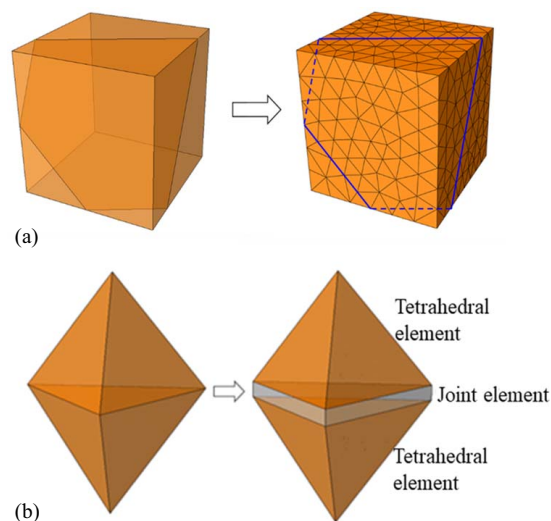


Fig. 1. (a) Mesh generation for fractured rock mass; and (b) connection of two tetrahedral elements with a joint element.

normal opening o_p is the normal displacement at which the joint element's stress approaches the tensile strength in a state of pure tension. The critical tangential displacement s_p is the tangential displacement at which the joint element's stress approaches the shear strength in a state of pure shear. According to the previous studies (Lisjak et al. 2013; Munjiza et al. 1999), a damage variable D of the joint element can be defined by the following equations:

$$D = \begin{cases} 0, & o < o_p \text{ and } s < s_p \\ \frac{o - o_p}{o_r - o_p}, & o \geq o_p \text{ and } s < s_p \\ \frac{s - s_p}{s_r - s_p}, & o < o_p \text{ and } s \geq s_p \\ \sqrt{\left(\frac{o - o_p}{o_r - o_p}\right)^2 + \left(\frac{s - s_p}{s_r - s_p}\right)^2}, & o \geq o_p \text{ and } s \geq s_p \end{cases} \quad (1)$$

where o_r = normal opening displacement of the joint element when it is just in the state of pure tensile failure; and s_r = the tangential opening displacement of the joint element when it is just in pure tangential failure. If the computed D from Eq. (1) is greater than 1, the value should be assumed to be 1.

The reduction coefficient of tangential stress and normal stress $f(D)$ of the joint element is calculated by using D (Munjiza 2004):

$$f(D) = \left(1 - \frac{a + b - 1}{a + b} e^{(D(a+b)/(a+b)(1-a-b))}\right) (a(1-D) + b(1-D)^c) \quad (2)$$

where a , b , and c = experimental parameters that can be obtained by fitting the material under tension test curve, with $a = 0.63$, $b = 1.8$, and $c = 6.0$, and the actual fitting of one of the experimental curves can be seen in the literature (Munjiza et al. 1999). As a result, the normal and tangential stresses of the joint element can be written as follows (Lisjak et al. 2013; Munjiza et al. 1999; Yan and Jiao 2018; Yan et al. 2021c):

$$\sigma = \begin{cases} \frac{2o}{o_p} f_t & \text{if } o < 0 \\ \left(\left[\frac{2o}{o_p} - \left(\frac{o}{o_p}\right)^2\right] f_t\right) f(D) & \text{if } 0 \leq o \leq o_p \\ f(D) f_t & \text{if } o > o_p \end{cases} \quad (3)$$

$$\tau = \begin{cases} \left[\frac{2|s|}{s_p} - \left(\frac{|s|}{s_p}\right)^2\right] (-\sigma \tan(\phi) + c) f(D) & \text{if } 0 \leq |s| \leq s_p \\ (-\sigma \tan(\phi) + c) f(D) & \text{if } |s| > s_p \end{cases} \quad (4)$$

where σ = normal stress; τ = tangential stress; f_t = tensile strength; and c and ϕ = cohesion and internal friction angles of the joint element, respectively.

Note that the joint element parameters need to be determined by the tests of uniaxial compression and Brazilian splitting. The detailed calibration methods can be found in the literature (Tatone and Grasselli 2015). The same parameters are used for all joint elements after calibration.

3D FDEM Grouting Model

Basic Assumptions

The permeability of the rock matrix is very small compared with the fracture. Therefore, the 3D grouting model of this paper

assumes that the permeability of the rock matrix is zero and that grout flow exists only in fractures, that is, grout flow occurs only in broken joint elements in the 3D FDEM.

Grout Flow Model

Before performing 3D FDEM analysis, existing fractures in the solid are used as constraints to generate a tetrahedral finite-element mesh in the preprocessing model. The joint elements located on the natural fracture are marked as broken joint elements, and joint elements with bonding effects are inserted into the common triangular surfaces of adjacent tetrahedral elements to represent the continuum. Thus, the fracture surface is formed by a set of broken joint elements, that is, spatial triangular surfaces. The flow of grout in cracked rock masses is converted to flow in a set of triangular surfaces.

As shown in Fig. 2, the fracture surface is discretized into triangles. The common nodes between adjacent triangular surfaces, labeled as Nodes 1–9 in Fig. 2, constitute the nodes for the calculation of the grout flow. The pressure distribution on the fracture surface is represented by the grout pressure at these nodes. The pressure at any point on the fracture surface can be obtained by linear interpolation of the pressures at nodes. In the following, we introduce grout pressure calculation on the fracture surface based on the mesh shown in Fig. 2.

Taking $\Delta 123$ in Fig. 2 as an example, it is assumed that the grout pressures at Nodes 1, 2, and 3 are p_1 , p_2 , and p_3 , respectively. In a local coordinate, the x_1 -axis is along Edge 12, the x_2 -axis is orthogonal to x_1 and located within the triangular surface, and the x_3 -axis is perpendicular to the triangular surface following the right-hand rule. Let us assume that the grout pressure on the triangular surface follows linear distribution and the grout pressure gradient within the triangle surface is constant and can be written in a local coordinate as follows:

$$\frac{\partial p}{\partial x_i} = \frac{1}{A} \int_A \frac{\partial p}{\partial x_i} dA \quad (i = 1, 2) \quad (5)$$

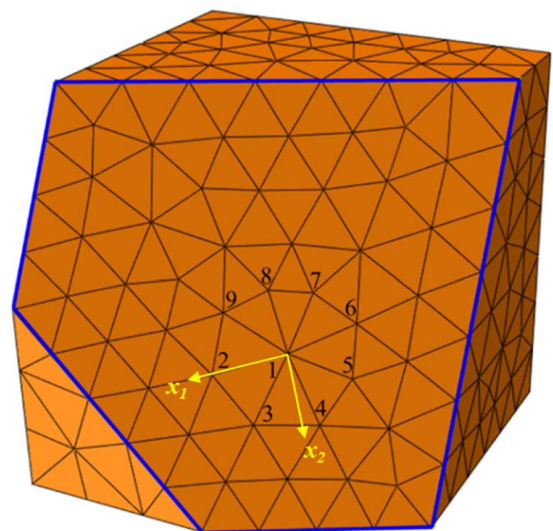


Fig. 2. Mesh of a fracture surface in the 3D FDEM.

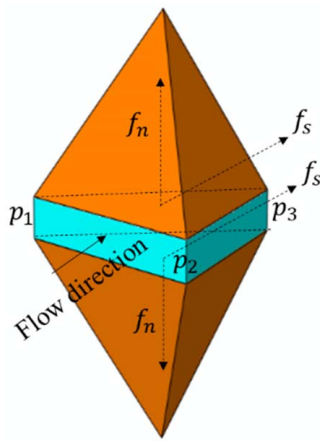


Fig. 3. Grout pressure in the broken joint element.

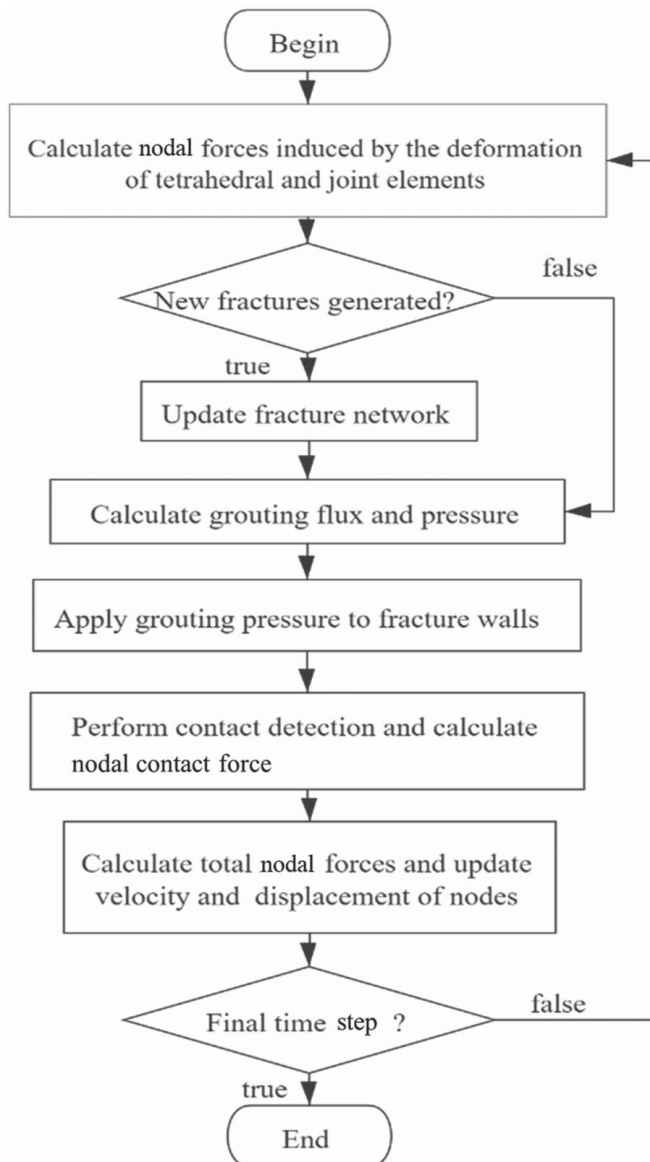


Fig. 4. Flowchart of the implementation in the 3D grouting model.

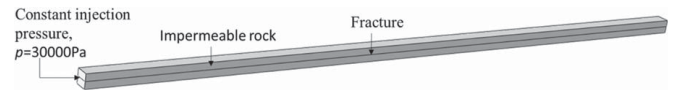


Fig. 5. Geometry and boundary conditions of the model.

Table 1. Parameters for model validation

Parameters	Value
Injection pressure, p (Pa)	30,000
Density of the grout, ρ (kg/m ³)	1,400
Viscosity of the grout, μ_0 (Pa·s)	0.04
Yield stress (for the Bingham fluid), τ_0 (Pa)	20
Aperture of the fracture, a (m)	5×10^{-3}

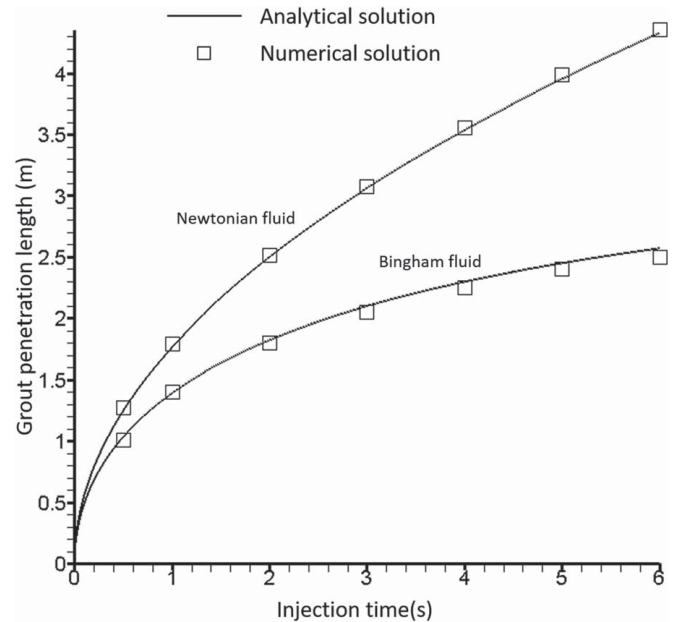


Fig. 6. Variation of penetration length with time for the numerical solution and the analytical solution.

According to the Gaussian divergence theorem, Eq. (1) can be written as follows:

$$\frac{\partial p}{\partial x_i} = \frac{1}{A} \int_s p n_i ds = \frac{1}{A} \sum_{m=1}^3 \bar{p}^m \epsilon_{ij} \Delta x_j^m \quad (6)$$

where A = triangular surface area; n_i = outer normal vector; $\bar{p}^m$ = average grout pressure at edge m ; Δx_j^m = difference in coordinate components between two edge m vertices (counted counterclockwise around the triangle); and ϵ_{ij} = two-dimensional permutation tensor, $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

The flux in the triangular surfaces is divided into the following three cases according to the types of grout (Nonveiller 2013; Yang et al. 2011).

If the grout is a Newtonian fluid, the flux of the grout in fractures is given by

$$q_i = -\frac{a^3}{12\mu} \frac{\partial p}{\partial x_i} \quad (7)$$

where q_i = flux along the i direction; μ = viscosity of the grout; a = averaged hydraulic aperture of the fracture; and p = grout pressure.

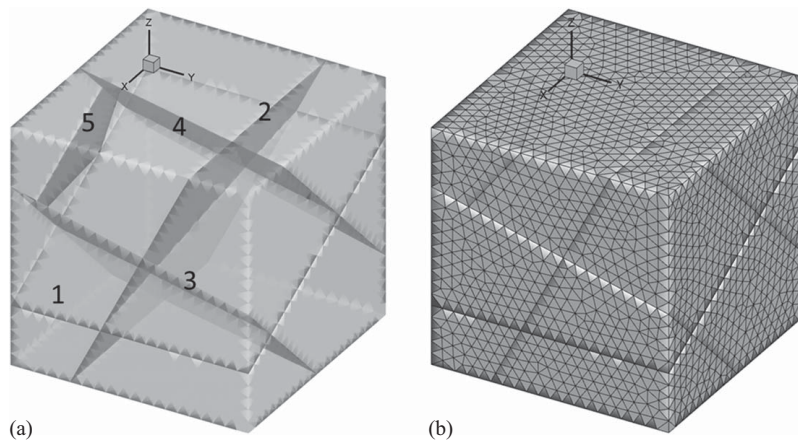


Fig. 7. (a) Three-dimensional model with a complex fracture network; and (b) the computational mesh.

If the grout is a Bingham fluid, the flux of the grout in fractures is given by

$$\begin{cases} q_i = -\frac{a^3}{12\mu} \frac{\partial p}{\partial x_i} \left(1 - \frac{3\tau_0}{|\text{grad}(p)|a} + \frac{4\tau_0^3}{|\text{grad}(p)|^3 a^3} \right) \\ \text{if } |\text{grad}(p)| > \frac{2\tau_0}{a} = 0 & \text{otherwise} \end{cases} \quad (8)$$

where τ_0 = yield stress of the grout. If the grout is a power-law fluid, the flux of the grout in fractures is given by

$$q_i = \frac{a^2}{2 + (1/n)} \left[\frac{\partial p}{\partial x_i} \frac{a}{2k} \right]^{1/n} \quad (9)$$

where n = behavioral index of the power-law fluid; and k = consistency index of the power-law fluid.

Taking Node 1 in Fig. 2 as an instance, the triangular surfaces that connect to it are $\Delta 123$, $\Delta 134$, $\Delta 145$, $\Delta 156$, $\Delta 178$, and $\Delta 189$. Then, the grout flow into Node 1 from the triangular surface $\Delta 123$ can be expressed by

$$Q_{\Delta 123 \rightarrow 1} = -\frac{q_i n_i^{(1)} L^{(1)}}{2} \quad (10)$$

where $n_i^{(1)}$ = outer normal vector of the edge opposite Node 1; and $L^{(1)}$ = length of the edge opposite Node 1. Similarly, the grout flow into Node 1 from other triangular surfaces that connect to Node 1 can also be obtained. Then, the total flow into Node 1 can be expressed by

$$Q_{\text{total} \rightarrow 1} = Q_{\Delta 123 \rightarrow 1} + Q_{\Delta 127 \rightarrow 1} + Q_{\Delta 176 \rightarrow 1} + Q_{\Delta 156 \rightarrow 1} + Q_{\Delta 145 \rightarrow 1} + Q_{\Delta 134 \rightarrow 1} \quad (11)$$

The grout pressure at the next time step of Node 1 can be obtained by

$$p_1^{t+\Delta t} = p_1^t + K_g Q_{\text{total} \rightarrow 1} \frac{\Delta t}{V^{t+\Delta t}} - K_g \frac{\Delta V}{V_m} \quad (12)$$

where p_1^t and $p_1^{t+\Delta t}$ = grout pressures of Node 1 at the current time step and the next time step, respectively; K_g = bulk modulus of the grout; $Q_{\text{total} \rightarrow 1}$ = total flow into Node 1; Δt = time step size; $\Delta V = V^{t+\Delta t} - V^t$; and $V_m = (V^{t+\Delta t} + V^t)/2$, where $V^{t+\Delta t}$ and V^t = node volumes (a half of the total volume of hydraulic fractures connected to this node) at the next and current time steps, respectively.

According to Eq. (12), if the nodal pressure is negative, it will be set to 0, which indicates that the grout flowing into the node is insufficient to fill the equivalent volume. Then, the saturation degree

Table 2. Model parameters for grouting

Model parameters	Value
Viscosity of the grout, μ_0 (Pa·s)	0.02
Yield stress of the grout, τ_0 (Pa)	15
Injection pressure, p (MPa)	1.5
Aperture, a (m)	7×10^{-5}

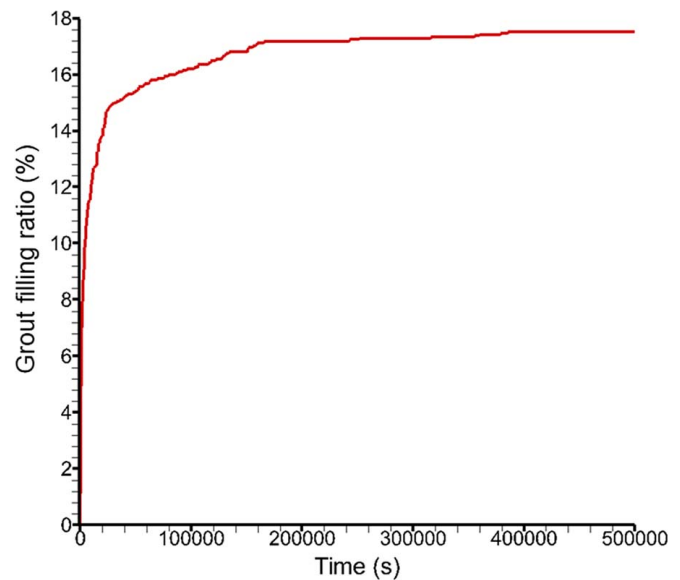


Fig. 8. Variation of the grout filling ratio with time.

of the node should be updated by

$$s^{t+\Delta t} = s^t + Q \frac{\Delta t}{V^t} - \frac{\Delta V}{V_m} \quad (13)$$

where $s^{t+\Delta t}$ and s^t = degree of node saturation at the following and the previous time step, respectively. The grout pressure of the node is set to zero if the saturation of node $s^{t+\Delta t} < 1$ at the following time step. The saturation is set to $s^{t+\Delta t} = 1$ if the saturation of node $s^{t+\Delta t} > 1$, and the node's grout pressure at the following time step is calculated using Eq. (12).

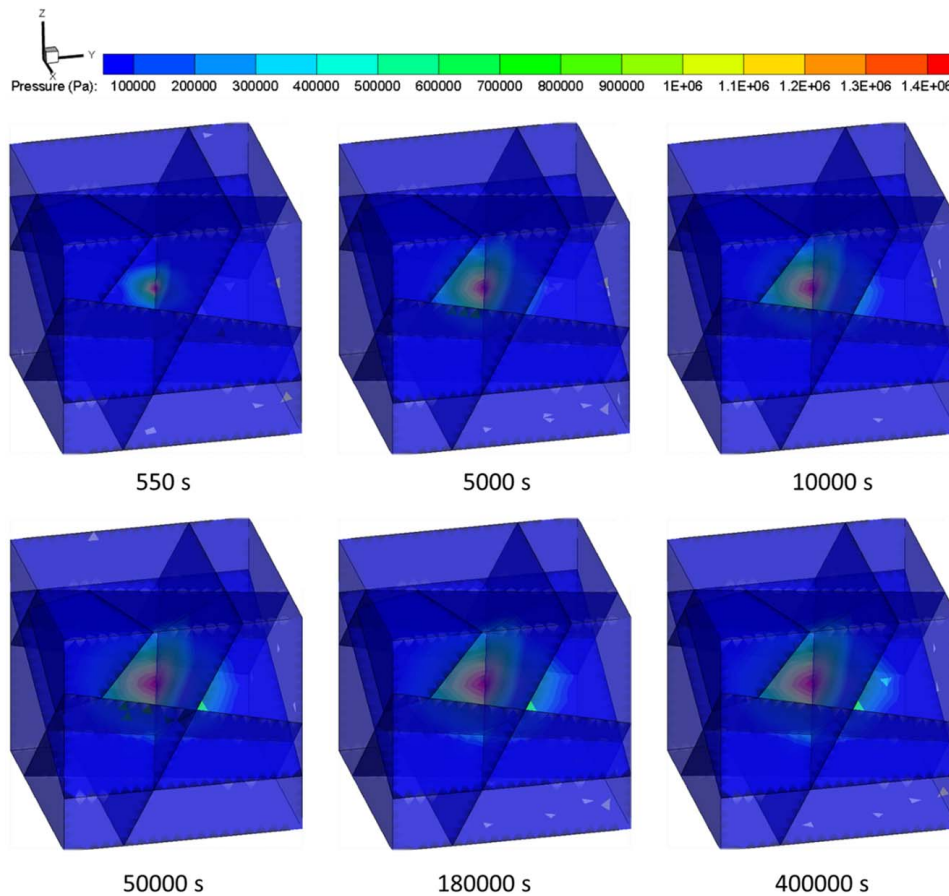


Fig. 9. Evolution of the grout pressure distribution with time.

The explicit integration strategy is used for the grouting flow calculation, the critical time step is defined as follows:

$$\Delta t_f = \min \left[\frac{V}{K_w \sum_i \frac{a_i^2}{12\mu}} \right] \quad (14)$$

where V = node volume, that is, a half of the total volume of hydraulic fractures connected to a node; a_i = the averaged hydraulic aperture of the fracture i ; and K_w and μ = bulk modulus and viscosity of the grout.

Interaction between Grout Flow and Deformation and Fracturing of Rock Masses

The following are the primary components of the interaction between grout flow and deformation and fracturing of the rock mass: (1) The grout flow exerts grout pressure and tangential viscous force on cracks, which may change the aperture of fractures in the rock mass, induce fracture propagation, or generate new fractures; and (2) On the other hand, the opening, closing, and propagation of fractures and the generation of new fractures affect the grout flow in fractures and changes the grout pressure.

Effect of Grout Flow on the Deformation and Fracturing of Rock Masses

According to the grout flow model in the “Grout Flow Model” section, the grout pressures at each node can be obtained and the two

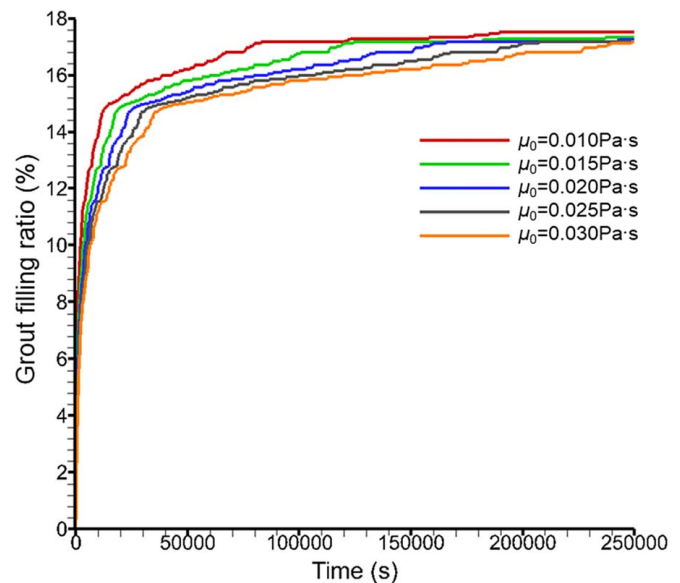


Fig. 10. Variation of the grout filling ratio over time for different viscosities.

triangular surfaces of the adjacent tetrahedral elements, that is, the broken joint elements are then applied. As shown in Fig. 3, the pressures at Nodes 1, 2, and 3 are denoted as p_1 , p_2 , and p_3 , respectively, resulting in a total force f_n acting on the triangular surface,

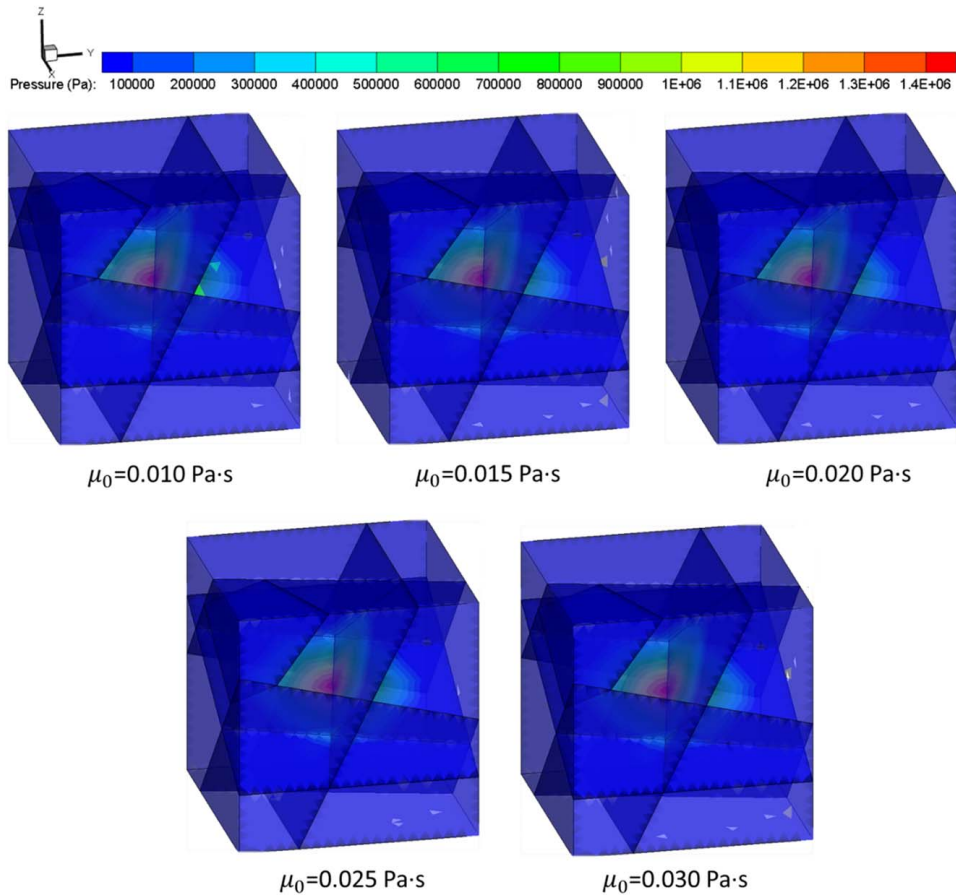


Fig. 11. Final pressure distribution of the grout for different viscosities.

given by

$$f_n = \frac{p_1 + p_2 + p_3}{3} A \quad (15)$$

In addition to grout pressure, the tangential viscous force of grout flow is also considered in the 3D grouting model. It is assumed that the grout flow in the broken joint element is a Poiseuille flow. The tangential viscous force exerted by the grout in the broken joint element is given by (Yan and Zheng 2017a)

$$\begin{aligned} f_{cx} &= A\tau_{f0x} = \frac{a}{2} A \frac{\partial p}{\partial x} \\ f_{cy} &= A\tau_{f0y} = \frac{a}{2} A \frac{\partial p}{\partial y} \end{aligned} \quad (16)$$

After the grout pressure and the tangential viscous force are calculated, they are uniformly distributed to three nodes of the broken joint element. Subsequently, the FDEM mechanical cracking calculation is carried out, and the effect of grout flow on the deformation and fracture of rock masses is considered.

Effect of the Deformation and Fracturing of Rock Masses on Grout Flow

The deformation of rock mass causes fractures to open or close. According to Eqs. (7)–(9), change in the fracture aperture affects the grout flow and finally affects the grout pressure according to Eq. (12). On the other hand, rock mass fracturing, that is, the propagation of existing fractures as well as initiation and propagation of

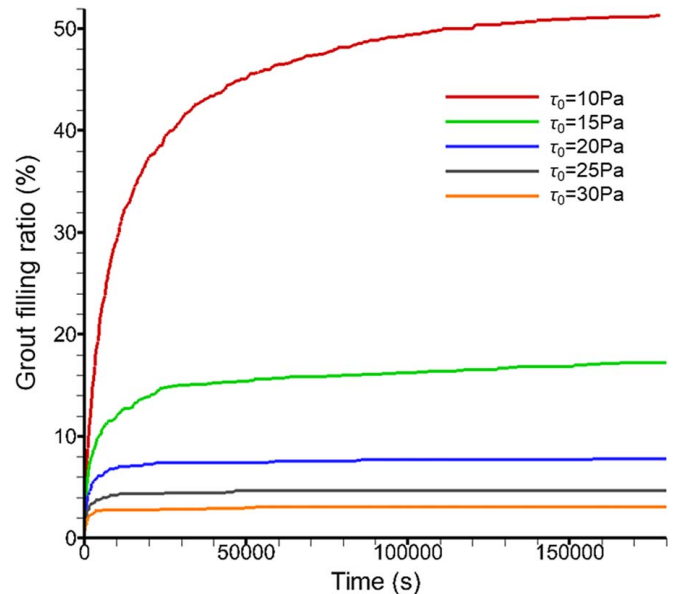


Fig. 12. Variation of the grout filling ratio over time for different yield stresses.

new fractures, also influences the grout flow. Through the breakage of joint elements during the initiation and propagation of fractures, the grout flow network changes, thereby affecting the grout flow rate and distribution of the grout pressure. In this way, the influence of rock mass deformation on grout flow and the effects of the

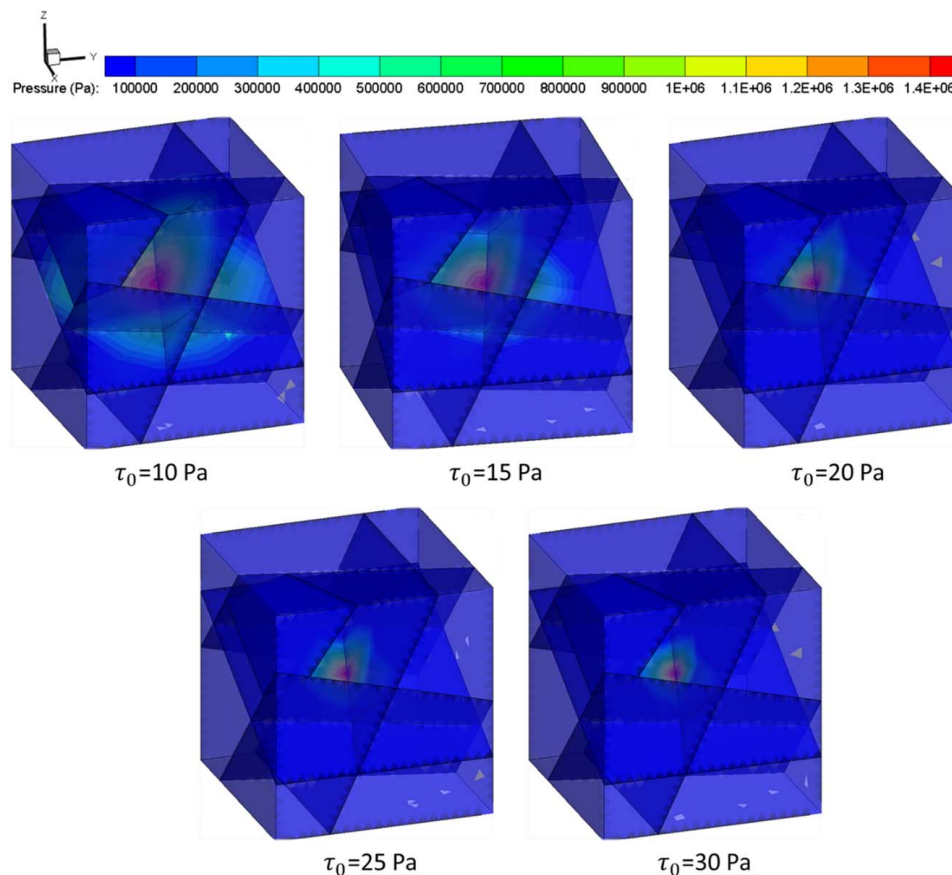


Fig. 13. Final grout pressure distribution for different yield stresses.

initiation and propagation of crack are considered in the proposed 3D grouting model of this paper.

Implementation of the 3D grouting model in this paper is illustrated in Fig. 4. First, according to the nodal displacements at the current time step, the nodal forces induced by the deformation of two different elements are calculated. Second, when the joint element breaking is detected and the fracture network is updated to include the generated new fractures. Third, the grout flow calculation is performed to update the nodal grout pressure and the tangential viscous force acting on the broken joint elements. Then, the grout pressure and the tangential viscous force are delivered to the tetrahedral elements joined by the joint elements as a surface load that is linearly spread out over the surface. Fourth, contact detection is performed between adjacent elements along with the fractures and the contact nodal force is calculated. Finally, total nodal forces are calculated, and all nodes' velocity and displacement are updated using Newton's second law. The calculation continues till the final time step.

Model Verification

In this section, a simple example is used to verify the 3D grouting model. As shown in Fig. 5, an impermeable rock matrix of $5 \times 0.1 \times 0.1$ m is divided by a single fracture into two pieces. Grout is injected from the left-hand side of the fracture under a constant inject pressure of 30 kPa, as shown in Fig. 5, and the sides of the model are prescribed as *no flow* boundaries. The rock matrix is discretized into 8,699 tetrahedral elements with a mesh size of 0.04 m. The grout fluid is considered as (1) a Newtonian fluid, or (2) a Bingham fluid. The 3D grouting model of this paper is used to solve the

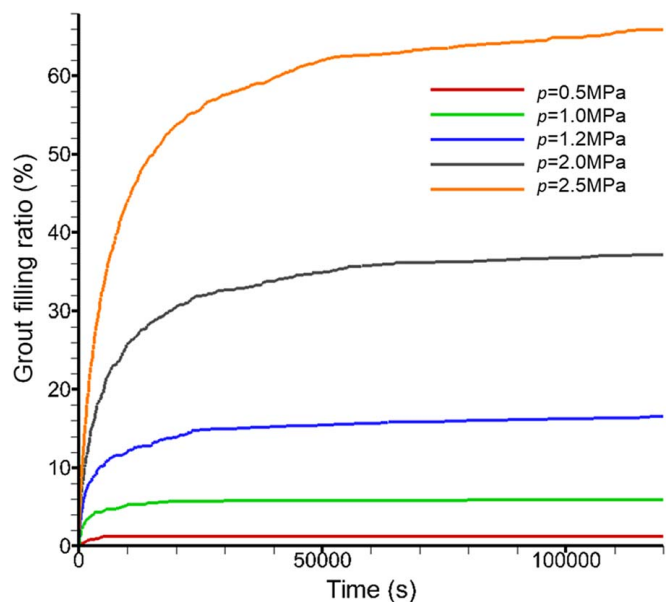


Fig. 14. Variation of the grout filling ratio over time for different injection pressures.

penetration length of the grout in these two cases. The parameters employed in the simulation are listed in Table 1. The viscosity of the grout is assumed to be constant.

Funehag and Gustafson (2008) obtained an analytical solution for the penetration length of a Newtonian fluid with a constant

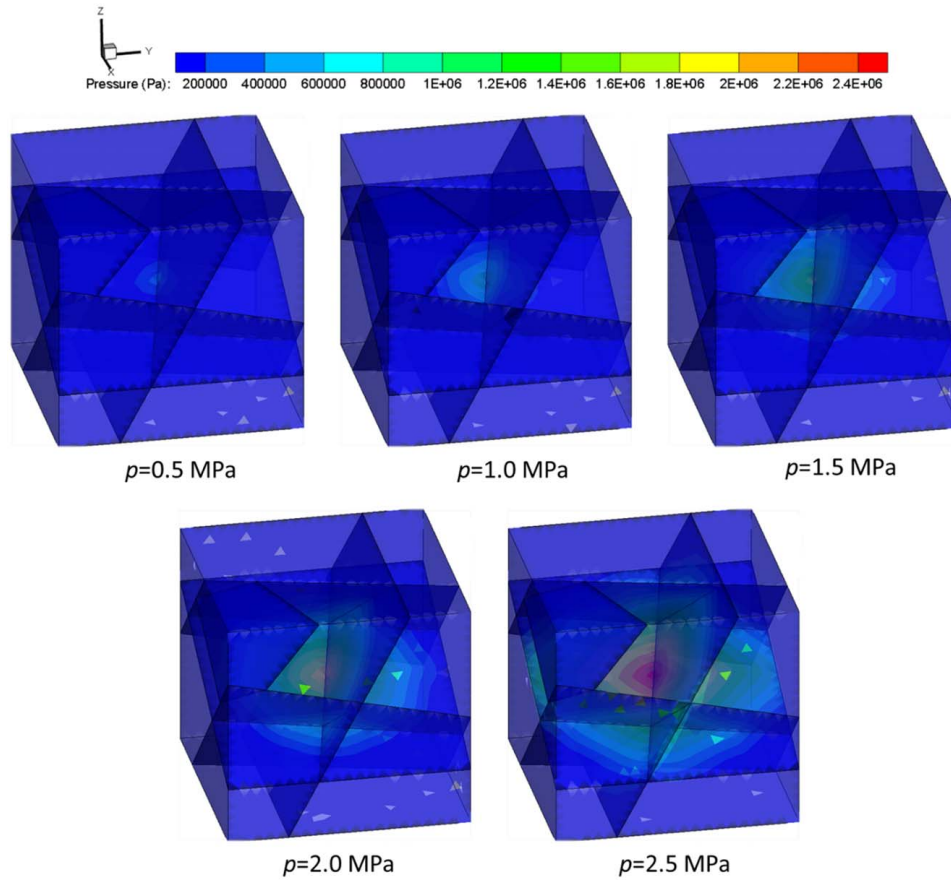


Fig. 15. Final pressure distribution of the grout for different injection pressures.

fluid viscosity:

$$I = a \sqrt{\frac{\Delta p t}{6\mu_0}} \quad (17)$$

where Δp = difference between the grout injection and the initial pore pressure; and t = grouting time.

Gustafson and Stille (2005) deduced an analytical solution for the penetration length of a Bingham fluid with a constant fluid viscosity:

$$I = I_D \cdot I_{\max} \quad (18)$$

$$t = t_D \cdot t_0 \quad (19)$$

$$I_D = \sqrt{\theta^2 + 4\theta} - \theta \quad (20)$$

$$t_0 = \frac{6\Delta p \cdot \mu_0}{\tau_0^2} \quad (21)$$

$$I_{\max} = \frac{\Delta p a}{2\tau_0} \quad (22)$$

where t_D and I_D = relative penetration time and relative penetration length, respectively; t_0 = characteristic grouting time; $I_{\max}$ = maximum grout penetration length; and θ = ratio influenced by t_D , which is given as follows for a one-dimensional flow:

$$\theta = \frac{t_D}{2(0.6 + t_D)} \quad (23)$$

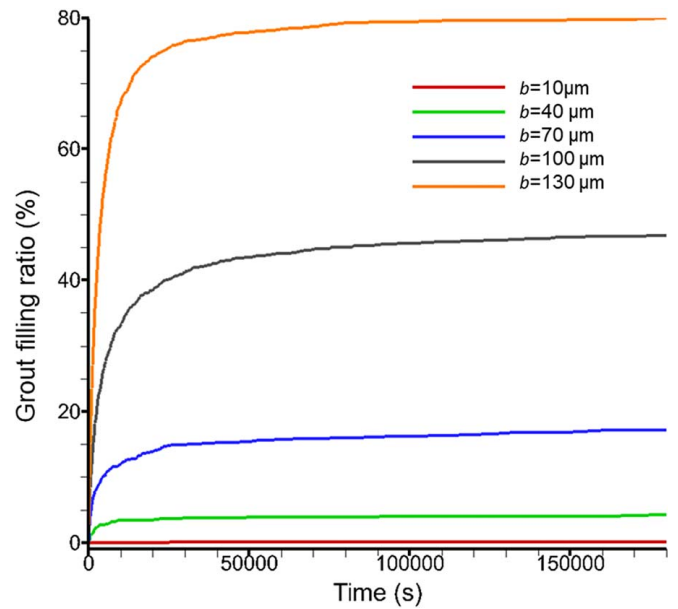


Fig. 16. Variation of the grout filling ratio over time for different apertures.

Fig. 6 demonstrates that the grout penetration length determined by the grouting model matches well with the analytical solution, indicating that the grouting model in this paper is accurate.

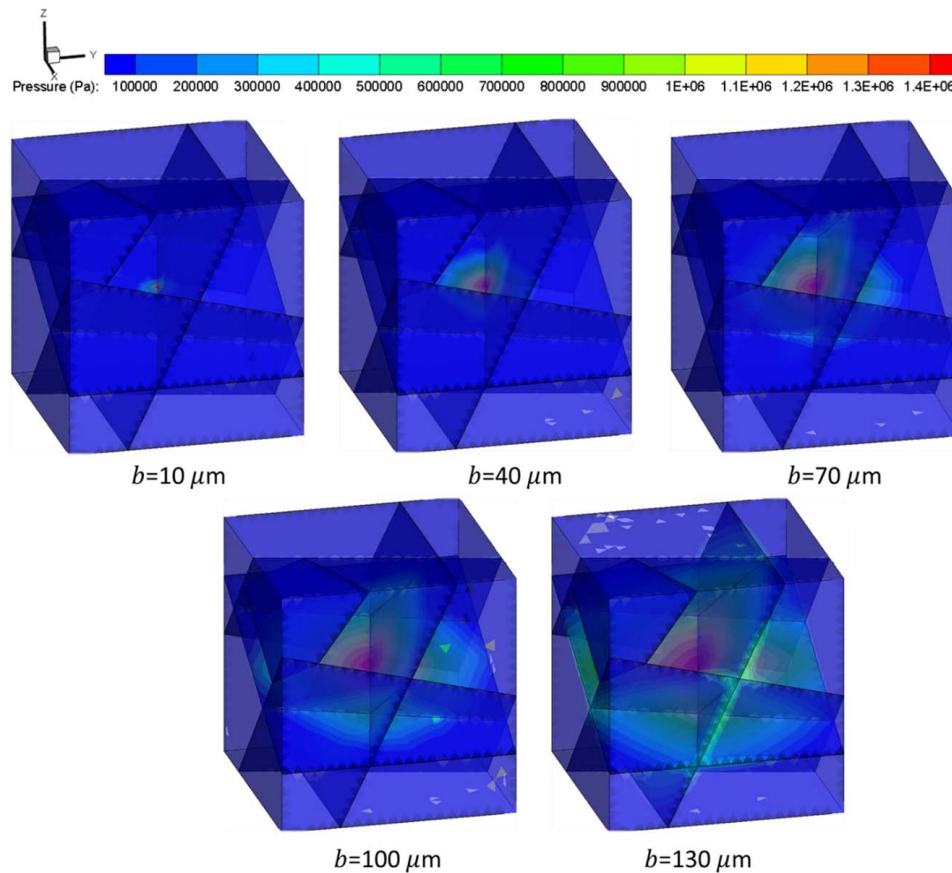


Fig. 17. Final pressure distribution of the grout for different apertures.

Grouting Procedure Simulation in Rock Masses with Complex Crack Networks

In this section, we use the proposed 3D grouting model to study the grouting procedure in a rock mass with a complex crack network. The rock matrix is a $10 \times 10 \times 10$ m cube, as shown in Fig. 7(a). Five fracture surfaces are randomly generated inside. Fractures 1 and 2 pass through the center of the model. Fractures 3 and 4 intersect with Fractures 1 and 2. Fracture 5 intersects at the far end of Fractures 3 and 4. The computational mesh is presented in Fig. 7(b) with 47,563 tetrahedral elements. Assuming that the grout is a Bingham fluid, the grout is injected into the fracture network from the center of the model under a certain pressure. The influence of the following parameters on grout migration is studied: (1) the material properties of the grout (viscosity, yield stress); (2) the grout injection pressure; and (3) the aperture of the fractures.

Table 2 presents the model parameters for a baseline case. These model parameters are varied to study their influences on the grouting process.

To quantitatively study the grouting effect, we define the grout filling ratio as follows:

$$\text{GFR} = \frac{L_g}{L_f} \quad (24)$$

where L_g = area of the grout-filled fractures; and L_f = area of total fracture. Due to the existence of yield stress, the grout does not flow until the shear stress of the grout is greater than the yield stress. Fig. 8 displays the variation in the grout filling ratio with the injection time for the baseline example. Within a relatively short period of time (approximately 10,000 s), the grout filling ratio reaches 12%,

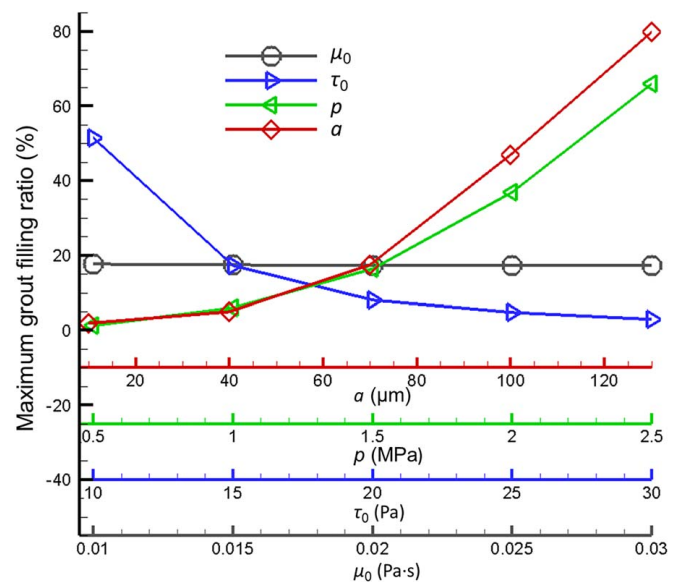


Fig. 18. Effects of grout viscosity, yield stress, injection pressure, and fracture aperture on the maximum grout filling ratio.

but the change of rate slows down gradually. At approximately 50,000 s, the grout filling ratio reaches 15.6%, and the final grout filling ratio is approximately 17.5%. Fig. 9 shows the grout pressure distribution in fractures at different times. The range of grout penetration increases rapidly in the first 10,000 s and then increases more slowly and eventually remains stable after 180,000 s.

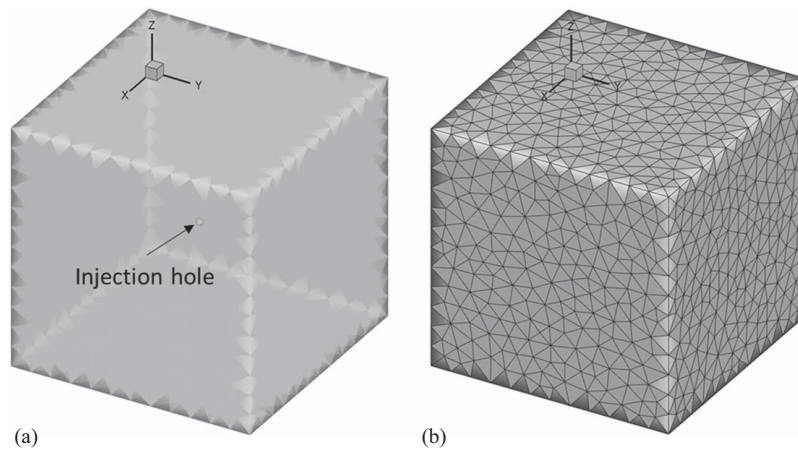


Fig. 19. (a) Model for simulating fracture grouting; and (b) the calculation mesh.

Influences of Grout Viscosity

The grout viscosity μ_0 is set to 0.01, 0.015, 0.02, 0.025, and 0.03 Pa·s, respectively, with the rest of the parameters keeping fixed from the baseline case. Fig. 10 depicts the variation in grout filling ratio over time for various viscosities. It can be seen that the greater the viscosity of the grout, the more slowly the grout flows, that is, the smaller the grout filling ratio at that time. However, the grout under different grout viscosities eventually reaches almost the same maximum grout filling ratio as time increases. The grout viscosity has little effect on the maximum grout filling ratio. Fig. 11 shows the final pressure distribution of the grout under different grout viscosities, which is almost the same, indicating that the final migration range of the grout under different grout viscosities is almost the same.

Influences of Grout Yield Stress

The grout yield stress τ_0 is taken to be 10, 15, 20, 25, and 30 Pa, respectively, to study the effect of yield stress on the grout filling ratio. The variation of grout filling ratio over time for yield stresses is shown in Fig. 12. It can be seen that as the grout yield stress increases, the maximum grout filling ratio decreases, and the grout migration range also decreases. Moreover, when the grout yield stress increases linearly, the maximum grout filling ratio decreases nonlinearly. When the grout yield stress increases from 10 to 20 Pa, the maximum grout filling ratio decreases from 51% to 8%. However, when the grout yield stress increases from 20 to 30 Pa, the maximum grout filling ratio decreases from 8% to 3%. Fig. 13 shows the final distribution of the grout under different grout yield stresses. It can be seen that the larger the grout yield stress is, the smaller the final distribution range of the grout.

Influences of Injection Pressure

We set the injection pressure p to 0.5, 1.0, 1.5, 2.0, and 2.5 MPa, respectively, and kept other parameters unchanged to study the effect of injection pressure on the grout filling ratio. As shown in Fig. 14, the final grout filling ratio increases nonlinearly with the increase in injection pressure. The greater the injection pressure, the faster the maximum grout filling ratio increases. Meanwhile, a greater injection pressure requires a longer time for the grout to reach the final grout filling ratio. For example, when the injection pressure is 0.5 MPa, the time required to achieve a stable grout filling ratio is approximately 5,000 s, but the time is approximately

Table 3. In situ stress

Case	σ_x (MPa)	σ_y (MPa)	σ_z (MPa)
I	0.1	0.1	0.1
II	0.1	0.1	0.5
III	0.1	0.3	0.8

Table 4. Input parameters for fracture grouting

Parameters	Value
Tetrahedral element	
Density, ρ (kg/m ³)	2×10^3
Elastic modulus, E (GPa)	4
Poisson's ratio, ν	0.25
Joint element	
Tensile strength, f_t (MPa)	0.3
Cohesion, c (MPa)	15
Internal friction angle, φ (°)	30
Mode I fracture energy release rate, G_{fI} (J/m ²)	0.5
Mode II fracture energy release rate, G_{fII} (J/m ²)	10
Penalty parameter, p_f (GPa)	400
Initial aperture, a_0 (m)	6×10^{-5}
Minimum aperture, $a_{\min}$ (m)	5×10^{-5}
Maximum aperture, $a_{\max}$ (m)	8×10^{-5}
Bingham fluid	
Bulk modulus of fluid, K_w (GPa)	2.2
Viscosity of fluid, μ_0 (Pa·s)	0.001
Density of fluid, ρ_f (kg/m ³)	1.25×10^3
Yield stress, τ_0 (Pa)	5

14,000 s when the grouting pressure is 2.5 MPa. Fig. 15 shows the final distribution of the grout at different injection pressures. It can be seen that the greater the injection pressure is, the greater the final migration range of the grout. However, it is worth noting that even if the injection pressure is increased to 2.5 MPa, there is still no grout in Fracture 5, because it is not directly connected to the injection hole. This shows that to improve the overall performance and economic benefits of grouting, it is not enough to simply increase the injection pressure. Rationally arranging multiple grouting holes may increase the efficiency of grout filling at a lower injection pressure.

Influences of Fracture Aperture

The influence of fracture aperture a on the grout filling ratio is studied by taking the fracture aperture to be 10, 40, 70, 100, and

130 μm , respectively, and keeping other parameters unchanged. The variation in the grout filling ratio with time under different fracture apertures is shown in Fig. 16. It can be seen that as the fracture aperture increases, the maximum grout filling ratio is significantly improved. When the fracture aperture is 10 μm , the maximum grout filling ratio is approximately 0.08%, which indicates a very small grout migration range. When the fracture aperture is 130 μm , the maximum grout filling ratio rises to 80.07%. Therefore, the fracture aperture has a greater influence on the grout filling ratio. Fig. 17 shows the final distribution of the grout using different fracture apertures. It can be seen that when the fracture aperture is 130 μm , the grout in Fractures 1 and 2 has reached the boundary of the model (*no flow* boundary), but still, there is no grout distributed in the Fracture 5 because it intersects at the far end of Fractures 3 and 4 and is not directly connected to the injection hole.

Finally, the influences of grout viscosity μ_0 , yield stress τ_0 , injection pressure p , and fracture aperture a on the maximum grout filling ratio are summarized in Fig. 18.

Simulation of Fracture Grouting under Different In Situ Stress

In this section, a three-dimensional cube model is built to study the influence of in situ stress on fracture grouting. The model's dimension is $6 \times 6 \times 6$ m. In the center, there is a 0.1 m-radius injection hole, as illustrated in Fig. 19(a). Fig. 19(b) presents

the computational mesh, which contains 15,443 tetrahedral elements, and the mesh is refined around the injection hole. Displacements are fixed in the normal direction around the boundaries of the model. We study the geometry of the induced grouting fracture under different combinations of in situ stresses as listed in Table 3. The model parameters are listed in Table 4.

The initial fluid pressure of 0.1 MPa is applied to the injection hole, and the pressure is increased by 100 Pa in each step until it reaches 2 MPa. A time step of 1×10^{-7} s is used throughout the calculations. Note that the model cube contains no fracture at the beginning of the simulations. Fig. 20 shows the evolution of grouting-induced fracture under stress condition Case I. Since the components of in situ stress in x -, y -, and z -directions are the same (0.1 MPa), the propagation direction of cracks becomes random. Several cracks are initiated and extend out from the injection hole almost in various directions, and finally, some of them reach the model boundary.

The evolution of grouting-induced fracture geometry in Case II is shown in Fig. 21. The cracks mainly propagate from the injection hole along the z -direction because the component of in situ stress in the z -direction (0.5 MPa) is greater than that in the other two directions (0.1 MPa). In the end, two obvious fracture branches reach the model boundary.

Fig. 22 shows the evolution of grouting-induced fracture shapes in Case III. It is obvious that the crack propagates along the direction perpendicular to the minimum principal in situ stress ($\sigma_x = 0.1$ MPa), and no branch crack is formed. Fig. 23 compares the

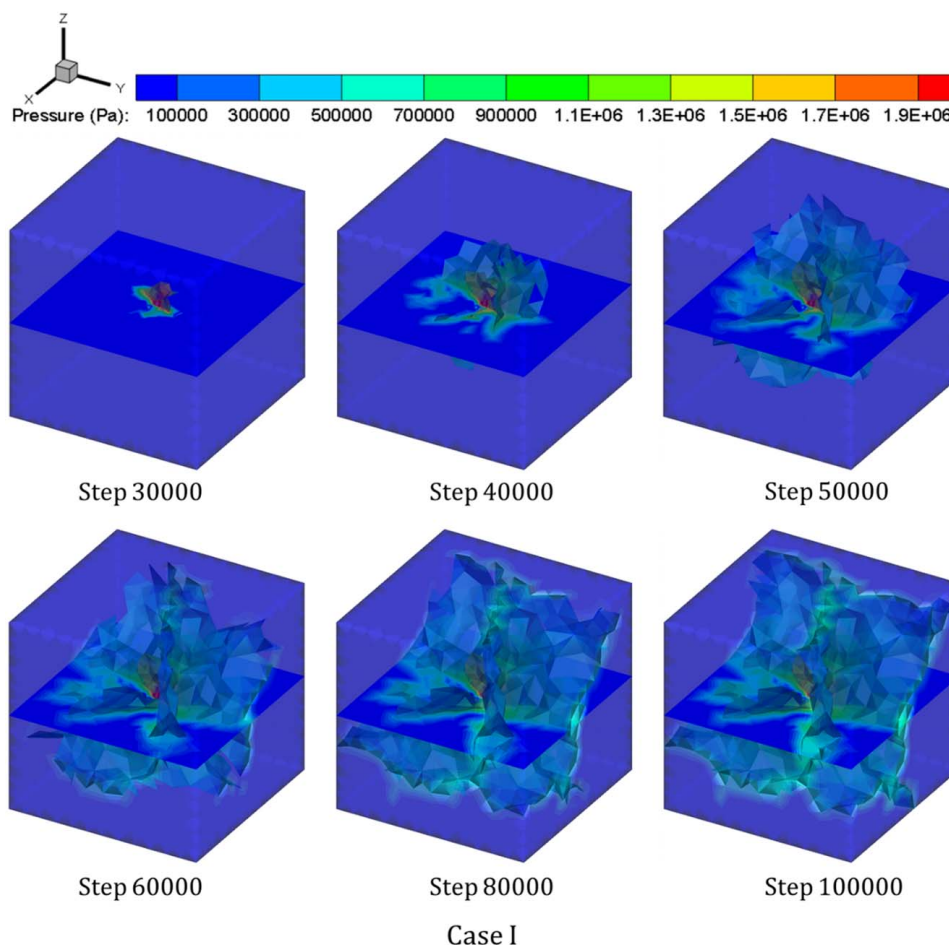


Fig. 20. Evolution of grouting-induced fracture for Case I.

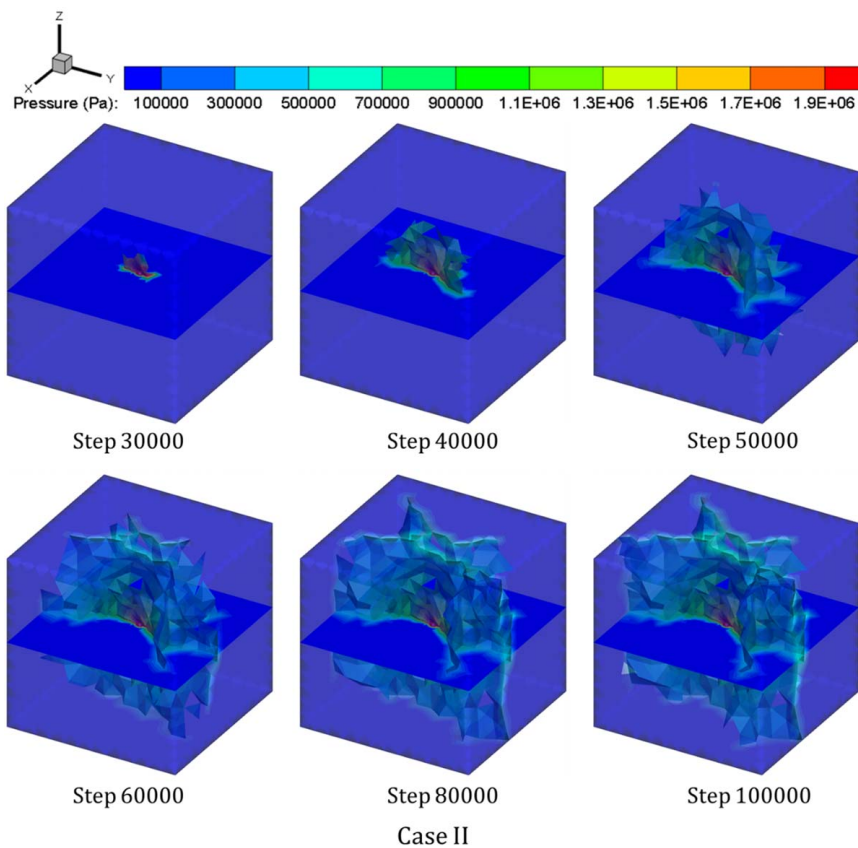


Fig. 21. Evolution of grouting-induced fracture for Case II.

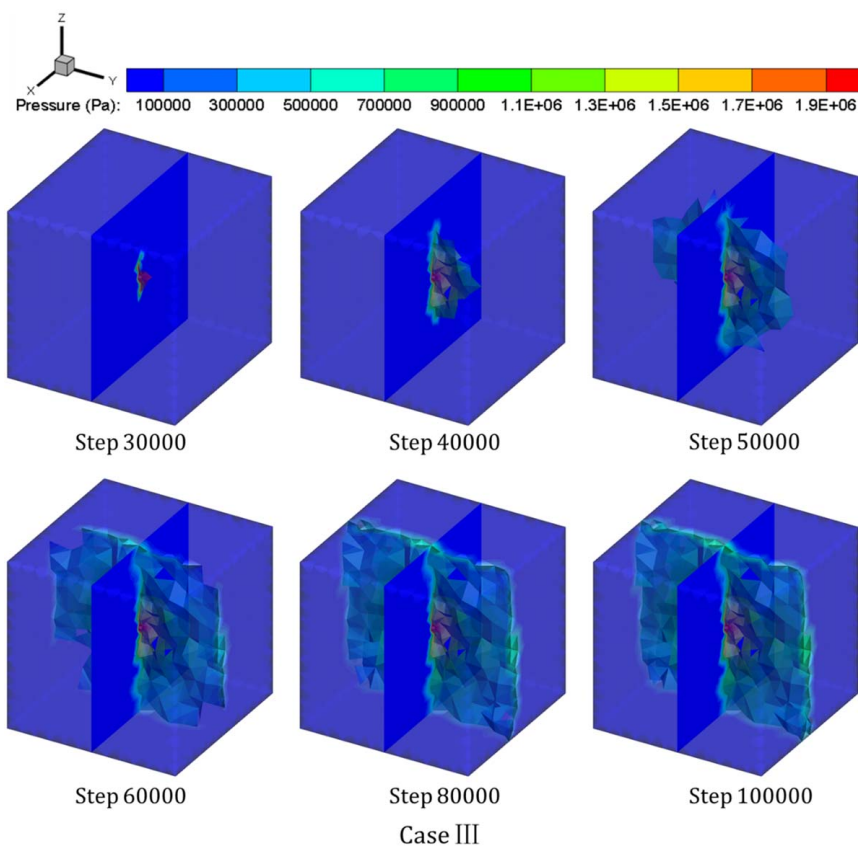


Fig. 22. Evolution of grouting-induced fracture for Case III.

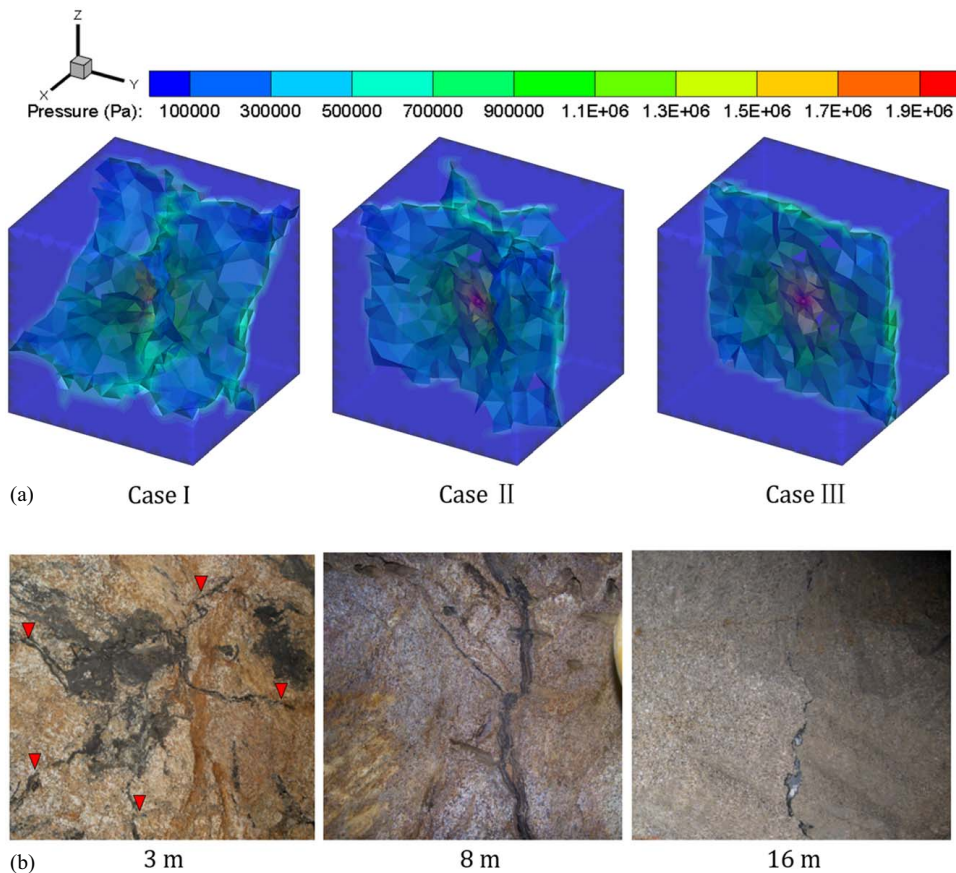


Fig. 23. (a) Grouting-induced fracture geometry under three different in situ stresses simulated by the FDEM; and (b) the grouting-induced fracture shapes at different depths of the in situ experiment.

final geometric shapes of grouting-induced fractures under three different in situ stresses simulated by FDEM [Fig. 23(a)], as against the in situ experimental observation of grouting-induced fractures at different depths [Fig. 23(b) (Chen et al. 2014)]. The numerical results are generally in good agreement with the in situ experimental results at similar stress conditions. It is interesting to note that the greater the in situ stress, the fewer branches and smaller areas of the induced grouting fractures.

Conclusions

In this study, an FDEM-based 3D grouting model for simulating the grouting process is developed. The 3D grouting model can simulate grout migration, pressure distribution, the interaction between grout and rock masses, rock deformation, and the initiation and propagation of cracks, which overcomes the drawbacks of previous grouting models.

By using the 3D grouting model, the main factors affecting the grouting penetration in complex fracture networks are investigated. The grout viscosity only affects the migration rate of the grout, and the smaller the viscosity is, the greater the migration rate. However, it has no effect on the grout's final migration range. The grout yield stress results in a smaller migration range of the grout. Moreover, the maximum grout filling ratio decreases nonlinearly when the grout yield stress increases. The greater the injection pressure is, the greater the final migration range. The greater the fracture aperture is the wider the grout migration range and the greater the grout migration rate. Studying these factors provides a certain reference value for improving the design of

grouting processes. Besides, the crack propagation behavior of fracture grouting under different in situ stresses is studied. The results show that the larger in situ stress results in a narrowly distributed grouting fracture with less branches and a smaller penetration area. These results are consistent with the in situ experimental results.

It should be noted that the viscosity of grout usually increases after injection till solidification. In addition, underground water usually exists in fractures. These conditions may affect the grout propagation and the final grouting effect. Therefore, in future work, we will consider the hardening process of the grout and the effect of underground water in the 3D grouting model.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

Acknowledgments

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